# React + TypeScript Roadmap (Basic → Advanced → Industry Level)

## Phase 1: TypeScript Fundamentals

### Topics

- What is TypeScript?
- Why TypeScript?
- TypeScript Compiler
- Type Inference
- Primitive Types
- string
- number
- boolean
- null
- undefined
- Arrays
- Objects
- Functions
- Optional Parameters
- Default Parameters
- Rest Parameters
- Type Aliases
- Interfaces
- Union Types
- Literal Types
- Enums
- any vs unknown
- void
- never

### Project

Student Management System

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## Phase 2: React + TypeScript Basics

### Topics

- Setting up React with TypeScript
- tsconfig basics
- Functional Components

- Typing Props
- Typing Children
- Default Props
- Component Composition
- Event Handling Types

### Project

Profile Card App

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## Phase 3: React State Management with TypeScript

### Topics

- useState
- Typing State
- Complex State
- Arrays in State
- Objects in State
- Forms
- Controlled Components
- Uncontrolled Components

### Project

Form Validation App

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## Phase 4: React Hooks with TypeScript

### Topics

- useEffect
- Dependency Array
- Cleanup Functions
- useRef
- DOM References
- useMemo
- useCallback
- React.memo
- Custom Hooks
- useReducer

### Project

Counter Dashboard

Features:

- Counter
- Timer
- Search
- Expensive Calculation
- Memoization

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## Phase 5: API Handling

### Topics

- Fetch API
- Async/Await
- Promise Handling
- Loading State
- Error State
- Data Mapping
- Custom API Hooks
- Axios Basics

### Project

Weather App (TypeScript)

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## Phase 6: Advanced TypeScript

### Topics

- Generics
- Generic Functions
- Generic Components
- Utility Types
- Partial
- Pick
- Omit
- Record
- Readonly
- Type Guards
- Narrowing
- Optional Chaining
- Nullish Coalescing

### Project

Reusable Data Table

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## Phase 7: Context API + TypeScript

### Topics

- createContext
- useContext
- Context Provider
- Custom Context Hooks
- Context Typing
- Context Architecture

### Project

Context Todo App (TypeScript)

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## Phase 8: React Router

### Topics

- BrowserRouter
- Routes
- Route
- Link
- NavLink
- useNavigate
- useParams
- Nested Routes
- Protected Routes

### Project

Multi-Page Blog UI

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## Phase 9: Redux Toolkit + TypeScript

### Topics

- Redux Fundamentals
- Store

- Slice
- Actions
- Reducers
- Dispatch
- Selectors
- Typed Store
- RootState
- AppDispatch
- Typed Hooks

### Project

Expense Tracker

Features:

- Add Expense
- Delete Expense
- Categories
- Total Expenses
- Filters

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## Phase 10: React Query (TanStack Query)

### Topics

- Why React Query?
- useQuery
- useMutation
- Query Keys
- Caching
- Refetching
- Optimistic Updates

### Project

GitHub Profile Finder (React Query)

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## Phase 11: Forms (Industry Standard)

### Topics

- React Hook Form
- Validation
- Zod

- Schema Validation
- Error Messages
- Form Architecture

### Project

Registration & Login Form

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## Phase 12: RTK Query

### Topics

- API Slice
- Queries
- Mutations
- Caching
- Tags
- Invalidation

### Project

Blog Dashboard

CRUD Operations:

- Create Post
- Read Post
- Update Post
- Delete Post

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## Phase 13: Performance Optimization

### Topics

- React.memo
- useMemo
- useCallback
- Code Splitting
- Lazy Loading
- Suspense
- Error Boundaries

### Project

Large Product Catalog

## Phase 14: Industry React Patterns

### Topics

- Folder Structure
- Feature-Based Architecture
- Absolute Imports
- Environment Variables
- Reusable Components
- Custom Hooks Architecture
- Service Layer
- Error Handling
- Loading States
- Empty States

### Project

Production-Level Dashboard

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## Phase 15: Advanced React

### Topics

- Portals
- Higher Order Components
- Render Props
- Compound Components
- React Strict Mode
- React Fiber
- Virtual DOM
- Reconciliation
- React Lifecycle
- Error Boundaries

### Project

Component Library

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# Week 3 (Optional but Highly Recommended)

## Authentication

- JWT

- Access Token
- Refresh Token
- Protected Routes

Project: Authentication System

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## Testing

- Jest
- React Testing Library
- Unit Testing
- Integration Testing

Project: Test Existing Todo App

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## Next.js

- App Router
- Server Components
- Client Components
- SSR
- SSG

Project: Personal Portfolio

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# Major Project 1

## Full Stack Expense Tracker

Tech:

- React
- TypeScript
- Redux Toolkit
- RTK Query
- Node.js
- Express
- MongoDB

Features:

- Authentication
- Expense CRUD

- Categories
- Charts
- Reports
- Dashboard

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# Major Project 2

## Full Stack Blog Platform

Tech:

- React
- TypeScript
- Redux Toolkit
- RTK Query
- React Hook Form
- Zod
- Node.js
- Express
- MongoDB

Features:

- Authentication
- Create Blog
- Update Blog
- Delete Blog
- Comments
- Likes
- User Profiles
- Admin Panel